

Raghav Senthil Kumar

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Designed a *Python (FastAPI)* data pipeline that prepares, transforms and delivers **600K+ debate-evidence documents** via *Elasticsearch*, returning results in **under 1 second** for a high school debate research SaaS platform
- Contributed to data quality activities with a de-duplication algorithm that flagged **30%** of documents as identical
- Supported data validation with a text-retrieval algorithm that checks source URLs and filters outliers using **IQR**
- Delivered cloud-hosted data search on *AWS* as part of an *Agile* team of **10 engineers**, serving **10K+ site visitors**

Product Development Intern | Jetson

January 2025 – February 2025

- Assisted with testing and validating **3 experimental features** through **7 iterations of UI/UX testing**
- Turned test results into actionable insights: a **5-page report** with strategies to optimize user engagement
- Validated a redesigned *Figma* onboarding flow through **A/B testing** with 30 users, cutting it from **12 to 5 pages**

Data Analytics Intern | STEM E

June 2024 – July 2024

- Processed and visualized **200+ TikTok posts** from **3 years** in *Python (pandas)*, *Power BI* and *Microsoft Excel*
- Identified a trend in the data: “science fun facts” posts drove **69.3%** of views from only **7%** of all posted content
- Generated an insight the Executive Director adopted as a “fun facts” strategy, lifting engagement **24%** (8K+ views)

PROJECTS

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Built a data integration pipeline that ingests papers via *OpenAlex* and backfills metadata from *Semantic Scholar*
- Wrote *SQL* queries on a relational *PostgreSQL* database to retrieve papers by field, year and citation count
- Delivered data visualization of a paper’s outbound citations across **26 research fields** in a *React Flow* graph
- Deployed a **RAG agent** (*Claude API*, *Supernemory*) for generative AI answers from session-scoped vector documents

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Prepared clean data from **15,000** truck-listing images by de-duplicating listings and filtering non-USD prices
- Transformed truck photos into structured condition, brand, model and year data with a *Claude Sonnet 4.6* agent
- Tested and validated the extraction, fixing each wrong prediction until **10 consecutive images** were right
- Trained a *quantile regression* pricing model on extracted data and validated **87% accuracy** on **30 samples**

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Built a *Python FastAPI* back end that ingests live water, waste and energy data from all **3 edge AI tools**
- Applied machine learning (**4 Random Forest models**) to turn data into actionable steps via a *Gemma Voice LLM*
- Created **3 edge AI tools** that collect data on a school’s energy, water and food consumption to support analytics
- Directed a team of 5 from ideation and development to an AI data product. **USAII Global Hackathon Finalist**

EcoBin | [Research Paper](#) | [Source Code](#)

March 2026 – May 2026

- Built an automated synthetic data pipeline (*U2-Net* segmentation, alpha-compositing) that generates **9K** images
- Trained and validated machine learning classifier (*EfficientNetV2-S*) on **15K+ images** to reach **96.13% accuracy**
- Documented the data and the machine learning methodology in an *arXiv* paper, releasing **2 datasets** on *Kaggle*
- Sorted waste into 4 disposal pathways with a two-stage AI classifier. **2nd at Arizona State Science Fair**

SKILLS

Programming Languages: Python, SQL, Java, C++, JavaScript, HTML/CSS

Frameworks/Platforms: PostgreSQL, MySQL, Supabase, AWS, Elasticsearch, FastAPI, Google Cloud, Flask, React

AI/ML: PyTorch, scikit-learn, TensorFlow, Keras, Hugging Face, Claude API, OpenCV, Gemma 4, Grad-CAM

Data & Analytics: pandas, NumPy, Power BI, Microsoft Excel, A/B Testing, Hootsuite

Concepts/Practices: Data Preparation, Data Transformation, Data Visualization, Databases, Generative AI